

Noah Cylich

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EDUCATION

University of California, Los Angeles

BS Computer Science, Minor in Mathematics. GPA 3.91, Dean's Honors

Los Angeles, CA

Expected December 2026

PROFESSIONAL EXPERIENCE

Cactus Compute

ML Engineer & Researcher

Los Angeles, CA

November 2025 – Present

- Implemented Gemma 4 end to end and shipped it fully multimodal on launch day, the only external provider supporting every modality, including the precision fixes that let the audio encoder run on the Apple Neural Engine
- 3rd largest contributor across 53 merged PRs to an open-source on-device LLM inference engine
- Optimized ARM decode kernels, making small matmuls 3x faster, lifting Gemma 4 E2B decode by 50%, alongside a hybrid INT8/FP16 attention kernel that doubled attention throughput during decode at unchanged accuracy
- Built KeyDiff KV-cache compaction and native chunked prefill, making 8k+ token context usable on mobile
- Co-authored the INT4 weight quantization kernels behind the engine's low-bit decode path
- Built the Go cloud backend that serves as the endpoint for every cloud hand-off request from the engine

Google

Software Engineer Intern, App Ads

Sunnyvale, CA

June 2026 – September 2026

- Built an agentic game image extraction and generation pipeline using Gemini and SAM 2
- Engineered a video labeling and analysis caching layer over the existing annotation system

Google

Software Engineer Intern, Database

Sunnyvale, CA

June 2025 – September 2025

- Architected an RPC-heavy automated presubmit testing framework in C++ for Napa, Google's exabyte-scale core database, validating every schema change behind services like AdSense
- Guaranteed data consistency for DeepMind queries by implementing repeatable read support for their Napa integration

Microsoft

Software Engineer Intern, Cybersecurity

Redmond, WA

June 2024 – September 2024

- Fixed a man-in-the-middle vulnerability in the Entra core authentication service
- Built an ASP.NET full-stack, LLM-powered chatbot with database plugins and secretless multi-tenant auth to assess client Authentication Resilience, which closed 99% of a Fortune 100 client's resiliency gap

RESEARCH

PLUS Lab, UCLA

LLM Researcher

Los Angeles, CA

September 2024 – Present

- **Temporal Mixture-of-Experts:** trained MoEs under a rolling-residency constraint that swaps at most one expert per token, so serving memory scales with active rather than total parameters
- Keeps 72–82% of the MoE-over-dense quality gain, and serves an 11B-scale model in llama.cpp at 5.1x less RAM for about a 25% slowdown
- With the mask lifted the router reuses 1.5–2x more of its previous experts than vanilla and each expert covers 2x more of the token stream (inverse Simpson), a mechanistic account of how the constraint makes the temporal experts contextual

PUBLICATIONS

- N. Cylich, K. Mosoyan, H. Ndubuaku, R. Shemet. "CLAWS: Calibration-Aware Activation Sparsity for Instruction-Tuned LLMs." *ICML 2026, AdaptFM Workshop*.
- N. Cylich. "Temporal Mixture-of-Experts: Serving Memory that Scales with Active Parameters." *In preparation*.
- S. Kumar, H. Ndubuaku, N. Cylich, K. Mosoyan, P. Sandhu, J. Lee, J. Mroz. "HiDRA: A Blazing Fast LM-Head Replacement." *ICLR 2026, Test-Time Updates (TTU) Workshop*.
- P. Sandhu, H. Ndubuaku, K. Mosoyan, S. Kumar, J. Lee, N. Cylich, J. Mroz. "TACE: Token-Aware Chunked Encoding." *ICLR 2026, NFAM Workshop*.
- H. Ndubuaku, K. Mosoyan, J. Mroz, N. Cylich, S. Kumar, P. Sandhu, R. Shemet, J. Lee. "A Controlled Study of Attention-Only Transformers." *arXiv preprint, 2026*.

LEADERSHIP

Bruin AI

Technical Director

Los Angeles, CA

Fall 2023 – 2025

- Led 5 engineers building an agentic system for Cymetrx that mapped CVEs to NIST 800-53 and HIPAA controls, working with the founder from scoping through a shipped demo
- Taught a quarter-long AI course for incoming members, from basic neural networks through transformers and NLP

TECHNICAL SKILLS

- Python, C++, PyTorch, JAX, Triton, CUDA, ARM NEON/SME2, LLMs, NLP, RL, C#, SQL, Docker, GCP, AWS